TECHNICAL CONCEPT PAPER

ROOT: A local architecture for room-level AC comfort

Technical concept paper

A developer-preview architecture and evaluation plan for bounded local room-comfort control.

ROOT-TCP-001 | D0.1

Developer preview | Issued 25 August 2026

Haptique Electronics Pvt. Ltd.

Document control

Table 1. Document-control register.

Field	Controlled value
Document ID	ROOT-TCP-001
Revision	D0.1
Publication status	Developer preview
Owner	Haptique Electronics Pvt. Ltd.
Technical reviewer	Not yet assigned
Issue date	25 August 2026
Last reviewed	25 August 2026
Evidence cutoff	25 August 2026
Hardware revision	Not yet assigned
Firmware revision	Not yet assigned

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Nomenclature and evidence status

Table 2. Nomenclature and evidence-status taxonomy.

Term or status	Meaning in this revision
AC	Air conditioner.
IR	Infrared command learning or transmission.
RSSI	Bluetooth received signal strength; a coarse and environment-dependent input.
ROOT	The proposed room-comfort controller described by this paper.
Cited background	A statement supported by an identified external source.
Implemented prototype behavior	A status reserved for behavior demonstrated by a revision-linked prototype record; no claim in D0.1 is assigned this status.
Hypothesis / design target	An intended behavior or bounded design thesis that remains unmeasured.
Planned evaluation	A claim framed as a protocol and acceptance decision to be completed later.

Abstract

A proposed local architecture for room-level AC comfort.

ROOT is a proposed controller for IR-operated split air conditioners. Its design thesis is to sense conditions closer to an occupied location, make bounded control decisions on the device, and transmit adjustments through the AC's existing infrared interface.

This paper describes the intended architecture, its control and privacy boundaries, and the evaluation methods required before product-performance statements can be made. It reports no measured product-performance result. All unmeasured ROOT behavior is classified as a hypothesis, design target, or planned evaluation.

1. The room-level problem

The occupied part of a room can differ from the return-air reference.

Split air conditioners typically regulate from a temperature reference near the indoor unit. Airflow, heat loads, room geometry, and occupant location can create non-uniform conditions that a single reference does not fully describe.

ROOT's design hypothesis is narrower: a sensor placed near a bed, couch, or desk may provide a more relevant control input for that location. The magnitude and usefulness of any difference remain dependent on the room and must be measured.

2. System architecture

A four-stage loop with a deliberately small control boundary.

The intended loop is: sense the room; estimate occupancy or context; decide locally; and transmit an infrared adjustment. Temperature, humidity, and a bounded presence signal are proposed inputs near the intended occupied location.

The proposed decision path uses bounded setpoints, bounded command rates, manual override, and a neutral fallback. An emitted infrared command is not treated as confirmed AC state. Conceptual architecture diagrams are not measured results.

1 | Sense	2 | Interpret	3 | Decide	4 | Transmit
Sample temperature, humidity, and bounded presence context.	Estimate local room context without inferring safety-critical state.	Apply bounded setpoints, command rates, override, and fallback rules.	Emit a learned IR command without assuming confirmed AC state.

Figure 1. Conceptual four-stage control loop. Conceptual architecture, not a measured result. Status: Hypothesis / design target. Claim ID: HT-01.

Signal path: environmental and bounded presence inputs -> local interpretation -> bounded local decision -> learned IR transmission. The proposed loop does not provide confirmed AC-state feedback.

Proposed control sequence

1. Sense temperature, humidity, and a bounded room-presence context near the occupied location.
2. Estimate context without treating coarse proximity as a safety-relevant or irreversible signal.
3. Decide locally within bounded setpoints, bounded command rates, manual override, and neutral fallback targets.
4. Transmit a learned IR adjustment without treating the emitted command as confirmed AC state.

3. Connectivity and local-control boundary

Local control is a scoped pathway, not a blanket privacy or uptime claim.

Bluetooth is a proposed configuration pathway. The intended remote-learning flow records supported IR commands from the existing AC remote; configuration storage, sensing, decision, and IR transmission are proposed device pathways. Exact permissions, pairing authorization, retained identifiers, storage, retention, deletion, sharing, and reset behavior require a reviewed data-flow design.

Offline operation must be evaluated after setup through sensing, decision, command, reboot, and reconnection scenarios. Bluetooth received signal strength varies with orientation, obstruction, radio conditions, and implementation; it is a coarse input and must not be the sole basis for a safety-relevant or irreversible action.

4. Sensing and infrared interface

Sensing and infrared are system elements awaiting integrated validation.

Temperature and humidity are proposed control inputs; part specifications and display resolution do not establish integrated system accuracy. Presence sensing is limited to room-control context and is not health or vital-sign monitoring.

IR learning is intended to record supported commands without implying broad compatibility. Infrared transmission reach depends on orientation, distance, line of sight, surfaces, emitter configuration, and the receiving AC. Illustrative product imagery is not evidence of final components, layout, construction, or production readiness.

Figure 2. Illustrative ROOT product cutaway. Illustrative image, not component or production evidence. Status: Hypothesis / design target. Claim ID: HT-03.

5. Placement and installation boundary

Placement changes what the device can sense and reach.

Tabletop, nightstand, couch-side, desk-side, and wall-mounted arrangements are placement concepts, not equivalent operating modes. Each changes sampled air, exposure to drafts or heat sources, sensor obstruction, mounting stability, occupant proximity, and the IR path to the AC.

Placement guidance must be tied to tested room and AC configurations. Users should retain access to manual AC controls when the recommended position cannot provide a reliable sensing or transmission path.

6. Claim and evidence register

The register below separates cited background, unmeasured design targets, and planned evaluations. It contains no claim assigned to Implemented prototype behavior.

Table 3. Claim and evidence register.

Claim ID	Status	Scope	Revision	Evidence ID	Evidence date	Control record
CB-01	Cited background	HVAC background and design motivation	Not applicable	REF-01	2026	Wording: Non-uniform conditions can arise in split-AC rooms, so a single return-air reference may not fully describe conditions near an occupant. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Not applicable Firmware revision: Not applicable
HT-01	Hypothesis / design target	Proposed four-stage room-control loop	Target revision not yet assigned	DESIGN-REVIEW-2026-08-25	25 August 2026	Wording: ROOT is designed to sense room context, decide locally within bounded controls, and transmit a learned IR adjustment. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned
HT-02	Hypothesis / design target	Proposed setup and local-control boundary	Target revision not yet assigned	REF-02; DESIGN-REVIEW-2026-08-25	25 August 2026	Wording: Bluetooth is a proposed setup or coarse-proximity pathway, while the intended core sensing, decision, and IR paths operate on the device after configuration. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned

Claim ID	Status	Scope	Revision	Evidence ID	Evidence date	Control record
HT-03	Hypothesis / design target	Proposed sensing and infrared interface	Target revision not yet assigned	DESIGN-REVIEW-2026-08-25	25 August 2026	Wording: Temperature, humidity, presence sensing, IR learning, and IR transmission require integrated validation before accuracy, reach, or compatibility claims are made. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned
HT-04	Hypothesis / design target	Placement concepts	Target revision not yet assigned	DESIGN-REVIEW-2026-08-25	25 August 2026	Wording: Placement affects sampled conditions, obstruction, airflow exposure, line of sight, IR reach, mounting stability, and presence-detection behavior. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned
PE-01	Planned evaluation	Climate sensing	Target revision not yet assigned	PROTOCOL-01	25 August 2026	Wording: Co-locate ROOT with a traceable reference logger and report bias, mean absolute error, repeatability, sampling interval, stabilization time, and uncertainty. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned
PE-02	Planned evaluation	Presence sensing	Target revision not yet assigned	PROTOCOL-02	25 August 2026	Wording: Compare presence detections with timestamped ground truth across occupied, unoccupied, stationary, moving, obstructed, placement, and false-trigger scenarios. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned

Claim ID	Status	Scope	Revision	Evidence ID	Evidence date	Control record
PE-03	Planned evaluation	IR interoperability	Target revision not yet assigned	PROTOCOL-03	25 August 2026	Wording: Test a declared stratified sample of AC brands, models, and commands over repeated trials, distances, angles, and line-of-sight conditions, reporting model-level successes and failures. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned
PE-04	Planned evaluation	Offline control	Target revision not yet assigned	PROTOCOL-04	25 August 2026	Wording: Disable Wi-Fi and internet after setup, then exercise sensing, decisions, commands, reboot, and reconnection for a stated duration. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned
PE-05	Planned evaluation	Setup	Target revision not yet assigned	PROTOCOL-05	25 August 2026	Wording: Observe first-time users completing a defined setup flow and report completion rate, timing distributions, assistance, retries, and failure reasons. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned
PE-06	Planned evaluation	Power behavior	Target revision not yet assigned	PROTOCOL-06	25 August 2026	Wording: Measure nominal and peak draw, supported input conditions, brownout and restart behavior, and recovery state using declared equipment. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned

Claim ID	Status	Scope	Revision	Evidence ID	Evidence date	Control record
PE-07	Planned evaluation	Room comfort stability	Target revision not yet assigned	PROTOCOL-07	25 August 2026	Wording: Use a randomized or counterbalanced baseline-versus-ROOT comparison with recorded conditions and predefined occupant-zone deviation, overshoot, cycling, and humidity outcomes. Owner: Haptique Electronics Pvt. Ltd. Review date: 25 August 2026 Superseded wording: None recorded Hardware revision: Target revision not yet assigned Firmware revision: Target revision not yet assigned

7. Evaluation methodology

Report protocols before results.

Every planned evaluation identifies the prototype and firmware revision, room or bench conditions, comparator and ground truth, sample interval, repeated trials, primary and secondary outcomes, acceptance criterion fixed before testing, exclusions and missing-data handling, uncertainty, and retained evidence artifact.

Future reports must include trial counts, negative results, protocol deviations, raw and derived data location, and analysis version. A simulator output is not a measured result.

Protocol-wide reporting rules

- Identify prototype hardware and firmware revisions, conditions, comparator, ground truth, sample interval, and repeated trials.
- Fix primary and secondary outcomes, acceptance criteria, exclusions, missing-data handling, and uncertainty before testing.
- Retain evidence artifacts and report trial counts, negative results, protocol deviations, data location, and analysis version.
- Keep simulator outputs distinct from measured results.

Planned evaluation IDs: PE-01, PE-02, PE-03, PE-04, PE-05, PE-06, and PE-07. Each acceptance rule will be pre-registered before testing; no threshold is approved in D0.1.

Table 4. Planned evaluation matrix.

ID	Evaluation	Conditions and comparator	Primary outputs	Acceptance rule
PE-01	Climate sensing	Relevant temperature and humidity conditions Comparator / ground truth: Traceable reference logger	bias; mean absolute error; repeatability; measurement uncertainty	Pre-registered before testing
PE-02	Presence sensing	Occupied, unoccupied, stationary, moving, obstructed, placement, and false-trigger scenarios Comparator / ground truth: Timestamped ground truth	sensitivity; specificity or false-positive rate; response latency	Pre-registered before testing

ID	Evaluation	Conditions and comparator	Primary outputs	Acceptance rule
PE-03	IR interoperability	Declared stratified AC sample; repeated distances, angles, and line-of-sight conditions Comparator / ground truth: Declared AC brand, model, and command set	model-level successes; model-level failures	Pre-registered before testing
PE-04	Offline control	Wi-Fi and internet disabled after setup for a stated duration Comparator / ground truth: Configured device before network loss	sensing, decision, command, reboot, and reconnection behavior	Pre-registered before testing
PE-05	Setup	First-time users completing one defined setup flow Comparator / ground truth: Observed user task completion	completion rate; median and range or interquartile range	Pre-registered before testing
PE-06	Power behavior	Declared supported input conditions and brownout/restart scenarios Comparator / ground truth: Declared measurement equipment	nominal draw; peak draw; brownout and restart behavior	Pre-registered before testing
PE-07	Room comfort stability	Recorded room geometry, placement, AC model, loads, outdoor conditions, and occupancy Comparator / ground truth: Randomized or counterbalanced baseline-versus-ROOT comparison	occupant-zone deviation; overshoot; cycling; humidity	Pre-registered before testing

8. Limitations, privacy, and safety boundary

A comfort controller with explicit limits.

ROOT is intended for general room-comfort control. It is not a medical device and is not intended to diagnose, prevent, monitor, predict, mitigate, or treat disease, injury, disability, a physiological condition, sleep, or a vital sign.

ROOT is not a safety-critical, emergency, or life-support controller. Users must retain the original AC remote and access to the AC's manual controls. Bounded setpoints, bounded command rates, visible fault indication, manual override, and a neutral fallback remain design targets until implemented and tested.

Validation must cover sensor faults, stale or out-of-range inputs, proximity loss, ambiguous or unsupported IR state, blocked line of sight, AC non-response, power interruption and restart, conflicting occupant preferences, mounting failure, and optional-connectivity loss. Compatibility and performance depend on AC hardware, room geometry, placement, obstruction, environmental conditions, power, firmware, and operating mode.

9. Company and document position

Haptique Electronics is developing ROOT.

Haptique Electronics Pvt. Ltd. is the company developing ROOT as a room-comfort product concept. ROOT is a product and brand, not a separate legal entity.

This revision intentionally omits third-party recognition and intellectual-property status. Those disclosures require verified identity, filing, scope, and approval records before inclusion. Developer contact: info@get-root.in.

References

References.

The cited background and Bluetooth measurement context are identified in the numbered reference register below.

Table 5. Numbered reference register.

ID	Author or organization	Title	Publication	Locator and access
REF-01	Haiyan Yan, Yawei Li, Thomas Parkinson, Stefano Schiavon, Hui Zhang, Rui Sun, Shengkai Zhao, Wei Zhao, Zhen Sun, Fangning Shi	Human thermal responses to non-uniform cooling from intermittently operated split air conditioners	Building and Environment (2026)	https://doi.org/10.1016/j.buildenv.2026.114379 Accessed 25 August 2026
REF-02	Bluetooth SIG	Bluetooth Low Energy Primer	Bluetooth SIG (Not dated)	https://www.bluetooth.com/bluetooth-le-primer/ Accessed 25 August 2026

Revision history

Revision history.

D0.1, issued 25 August 2026 by Haptique Electronics Pvt. Ltd.; reviewer: Not yet assigned; evidence cutoff: 25 August 2026. Initial developer-preview concept paper. No prototype performance result is reported.

Table 6. Revision history.

Revision	Issue date	Owner	Reviewer	Evidence cutoff	Summary
D0.1	25 August 2026	Haptique Electronics Pvt. Ltd.	Not yet assigned	25 August 2026	Initial developer-preview concept paper. No prototype performance result is reported.

Appendix A. Controlled disclosures

These controlled disclosures preserve the product's intended-use, safety, company, privacy, and omission boundaries.

Table 7. Controlled-disclosure register.

ID	Type	Exact wording	Source / approval	Owner	Review control
CD-01	Intended use and medical boundary	ROOT is intended for general room-comfort control. It is not a medical device and is not intended to diagnose, prevent, monitor, predict, mitigate, or treat disease, injury, disability, a physiological condition, sleep, or a vital sign.	WHITEPAPER-D0.1 / approved controlled disclosure	Haptique Electronics Pvt. Ltd.	Approved: 25 August 2026 Next review: Not yet assigned
CD-02	Safety role	ROOT is not a safety-critical, emergency, or life-support controller. Users must retain the original AC remote and access to the AC's manual controls.	WHITEPAPER-D0.1 / approved controlled disclosure	Haptique Electronics Pvt. Ltd.	Approved: 25 August 2026 Next review: Not yet assigned
CD-03	Company identity	Haptique Electronics Pvt. Ltd. is the company developing ROOT as a room-comfort product concept. ROOT is a product and brand, not a separate legal entity.	WHITEPAPER-D0.1 / company-supplied disclosure, not independently verified	Haptique Electronics Pvt. Ltd.	Approved: 25 August 2026 Next review: Not yet assigned
CD-04	Privacy and data boundary	A future Bluetooth proximity feature may process a paired identifier and received signal strength. Storage, retention, deletion, reset behavior, sharing, and location permissions are not yet assigned.	WHITEPAPER-D0.1 / approved controlled disclosure	Haptique Electronics Pvt. Ltd.	Approved: 25 August 2026 Next review: Not yet assigned
CD-05	Recognition and patent omission	This revision intentionally omits third-party recognition and patent-status statements. Those disclosures require verified identity, filing, scope, and approval records before inclusion.	WHITEPAPER-D0.1 / approved omission record	Haptique Electronics Pvt. Ltd.	Approved: 25 August 2026 Next review: Not yet assigned